8. Estimation of Parameters and Fitting of Probability Distributions

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SS26 MTH344: Mathematical Statistics

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Table of Contents

1. Introduction
2. Fitting the Poisson Distribution to Emissions of Alpha Particles
3. Parameter Estimation
4. The Method of Moments
5. The Method of Maximum Likelihood
6. The Bayesian Approach to Parameter Estimation
7. Efficiency and the Cramér-Rao Lower Bound
8. Sufficiency

Lecture Topics

M1. Review of Probability Theory

1. Probability
2. Random Variables
3. Joint Distributions
4. Expected Values

M2. Sampling Distributions and Estimation

5. Limit Theorems
6. Distributions Derived from the Normal Distribution
8. Estimation of Parameters and Fitting of Probability Distributions

M3. Hypothesis Testing

9. Testing Hypotheses and Assessing Goodness of Fit
11. Comparing Two Samples
12. The Analysis of Variance

Lecture Topics

- So far, we have studied the approximation of sampling distributions based on the Central Limit Theorem when n is large (Chapter 5), or based on the χ^2, t, or F distributions when n is small and the population is $\mathcal{N}(\mu, \sigma^2)$ (Chapter 6).
- This lecture introduces various parameter estimators to extend our analysis to non-normal populations. We will derive their asymptotic properties and compare them.
- See Chapter 8 in **Mathematical Statistics and Data Analysis [MSDA]** (Rice, 2007) for details. Selected examples and text excerpts are reproduced from [MSDA] for educational use only under fair use.
- This course presents content mostly following the textbooks, but also provides substantial unique and advanced material.

Table of Contents

1. Introduction
2. Fitting the Poisson Distribution to Emissions of Alpha Particles
3. Parameter Estimation
4. The Method of Moments
5. The Method of Maximum Likelihood
6. The Bayesian Approach to Parameter Estimation
7. Efficiency and the Cramér-Rao Lower Bound
8. Sufficiency

Fitting the Poisson Distribution to Emissions of Alpha Particles

n	Observed
0–2	18
3	28
4	56
5	105
6	126
7	146
8	164
9	161
10	123
11	101
12	74
13	53
14	23
15	15
16	9
17+	5
	1207

A table in Section 8.2. in [MSDA].

- The table shows the frequency of the number of alpha particles counted in 1,207 intervals. There is randomness in the number of observed particles.
- If the emission rate is constant and particles come from many independent sources (atoms), then the count follows

$$\mathrm{Bin}(n, p_n) \tag{1}$$

  where n is large and np_n is moderate.
- Therefore, we may use $\mathrm{Poi}(\lambda)$.
- **Q**: What is the optimal estimate for λ?

Fitting the Poisson Distribution to Emissions of Alpha Particles

n	Observed	Expected
0–2	18	12.2
3	28	27.0
4	56	56.5
5	105	94.9
6	126	132.7
7	146	159.1
8	164	166.9
9	161	155.6
10	123	130.6
11	101	99.7
12	74	69.7
13	53	45.0
14	23	27.0
15	15	15.1
16	9	7.9
17+	5	7.1
	1207	1207

A table in Section 8.2. in [MSDA].

- Given $E[X] = \lambda$ when $X \sim \mathrm{Poi}(\lambda)$, an intuitive choice would be:

$$\hat{\lambda} = \frac{1}{1207} \sum_{i=1}^{1207} x_i = 8.392. \tag{2}$$

- For example, when $X \sim \mathrm{Poi}(\hat{\lambda})$.

$$1207 \sum_{x=0}^{2} p(x; \hat{\lambda}) = 1207 \left(e^{-\hat{\lambda}} + \hat{\lambda} e^{-\hat{\lambda}} + \frac{\hat{\lambda}^2}{2} e^{-\hat{\lambda}} \right)$$
$$= 12.2.$$

- **Q**: How can we compare $\hat{\lambda}$ with other values of λ, and how can we find the optimal choice?

Table of Contents

1. Introduction
2. Fitting the Poisson Distribution to Emissions of Alpha Particles
3. Parameter Estimation
4. The Method of Moments
5. The Method of Maximum Likelihood
6. The Bayesian Approach to Parameter Estimation
7. Efficiency and the Cramér-Rao Lower Bound
8. Sufficiency

Parameter Estimation

- As in the previous example, we may apply a parametric distribution model (e.g., $\mathrm{Poi}(\lambda)$ or $\mathcal{N}(\mu, \sigma^2)$) to describe patterns in the data.
- In such cases, a key step is *parameter estimation*: choosing the values of the parameter θ (e.g., λ or $(\mu, \sigma^2)^T$).[1]
- A standard approach in statistics is to view the observed data $(x_i)_{i=1}^n$ as realizations of i.i.d. random variables $(X_i)_{i=1}^n$, each following $f_X(x;\theta)$ (or $p_X(x;\theta)$), and to construct an estimator as a function of the data, i.e., $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$.
- By doing so, we can analyze the stochastic behavior of our estimator $\hat{\theta}$ (its *sampling distribution*) and perform inference based on its variability (its *standard error*).

[1] Following tradition, we use θ to denote both a scalar and a vector of parameters.

Table of Contents

1. Introduction
2. Fitting the Poisson Distribution to Emissions of Alpha Particles
3. Parameter Estimation
4. The Method of Moments
5. The Method of Maximum Likelihood
6. The Bayesian Approach to Parameter Estimation
7. Efficiency and the Cramér-Rao Lower Bound
8. Sufficiency

The Method of Moments

- When the relationships between the parameters of a distribution and its moments are known, it is straightforward to construct estimators based on these relationships. This approach is referred to as *the Method of Moments*.
- The population r-th moment is defined as:

$$m_r = \mu_{X^r} = \mathrm{E}[X^r]. \tag{3}$$

- Given random samples $(X_i)_{i=1}^n$, the sample version (or *empirical estimate*) of the r-th moment is defined as:

$$\hat{m}_r = \hat{\mu}_{X^r} = \overline{X^r} = \frac{1}{n}\sum_{i=1}^{n} X_i^r. \tag{4}$$

- For example, when $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Poi}(\lambda)$, we know that $\lambda = \mathrm{E}[X] = m_1$. Given this, we may use $\hat{\lambda}^{\mathrm{MME}} = \hat{m}_1$, where MME stands for the <u>M</u>ethod of <u>M</u>oments <u>E</u>stimator.

The Method of Moments

- For a given target parameter, MMEs are not unique. For example, when $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Poi}(\lambda)$:

$$\mathrm{E}[X_1] = m_1 = \lambda \quad \text{and} \quad \mathrm{Var}[X_1] = m_2 - m_1^2 = \lambda. \tag{5}$$

  We can consider two different MMEs: $\hat{\lambda}_1^{\text{MME}} = \bar{X}$ and $\hat{\lambda}_2^{\text{MME}} = \overline{X^2} - \left(\bar{X}\right)^2$.
- Note that since λ parameterizes the PMF, all moments are functions of λ.
- Standard practice is to use relationships based on the lowest-order moments, as estimators derived from higher-order moments generally exhibit much higher variability (i.e., they are less efficient).

The Method of Moments

- **Example 8.4.B**: When $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, the first two moments can be expressed with the parameter $\theta = (\mu, \sigma^2)^T$ as follows:

$$m_1 = \mathrm{E}[X_1] = \mu \quad \text{and} \quad m_2 = \mathrm{Var}[X_1] + \mathrm{E}[X_1]^2 = \sigma^2 + \mu^2, \tag{6}$$

equivalently,

$$\mu = m_1 \quad \text{and} \quad \sigma^2 = m_2 - m_1^2. \tag{7}$$

- Based on this, we can construct MMEs as follows:

$$\hat{\theta}^{\mathrm{MME}} = \left(\bar{X}, \overline{X^2} - \left(\bar{X}\right)^2\right)^T. \tag{8}$$

The Method of Moments

- The Method of Moments is applicable in non-parametric settings as well.
- Suppose X has mean

$$\mu = \mathrm{E}[X]. \tag{9}$$

  We can still use $\hat{\mu}^{\mathrm{MME}} = \hat{m}_1$.
- For another example, suppose X has variance

$$\sigma^2 = \mathrm{Var}[X] = m_2 - m_1^2. \tag{10}$$

  We can use $\hat{\sigma^2}^{\mathrm{MME}} = \hat{m}_2 - \hat{m}_1^2$.

The Method of Moments

- We can extend this approach to the multivariate random vector.
- Suppose $(X, Y)^T$ has correlation coefficient:

$$\rho = \mathrm{Corr}[X, Y] = \frac{\mathrm{Cov}[X, Y]}{\sqrt{\mathrm{Var}[X]}\sqrt{\mathrm{Var}[Y]}} = \frac{\mathrm{E}[XY] - \mathrm{E}[X]\,\mathrm{E}[Y]}{\sqrt{\mathrm{E}[X^2] - (\mathrm{E}[X])^2}\sqrt{\mathrm{E}[Y^2] - (\mathrm{E}[Y])^2}}. \tag{11}$$

Based on this relationship, we can use the estimator:

$$\hat{\rho}^{\mathrm{MME}} = \frac{\overline{XY} - \bar{X}\,\bar{Y}}{\sqrt{\overline{X^2} - \left(\bar{X}\right)^2}\sqrt{\overline{Y^2} - \left(\bar{Y}\right)^2}}, \tag{12}$$

provided that the denominator is non-zero.

Sampling Distributions of MMEs

- Again, as functions of random samples $(X_i)_{i=1}^n$, estimators are random quantities that follow a sampling distribution.
- **Example 8.4.A**: When $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Poi}(\lambda_0)$, where λ_0 denotes the true underlying parameter value, and we estimate λ_0 under the Poisson assumption,[2] we can derive the distribution of:

$$\hat{\lambda}^{\text{MME}} = \hat{m}_1 = \bar{X}. \tag{13}$$

- Since $\sum_{i=1}^n X_i \sim \mathrm{Poi}(n\lambda_0)$, the PMF of $\hat{\lambda}^{\text{MME}}$ is given by:

$$P\left(\hat{\lambda}^{\text{MME}} = v\right) = \frac{(n\lambda_0)^{nv}}{(nv)!} e^{-n\lambda_0}, \quad v \in \{0, 1/n, 2/n, \dots\}. \tag{14}$$

Based on this, $\mathrm{E}_{\boldsymbol{X}}[\hat{\lambda}^{\text{MME}}] = \lambda_0$ and $\mathrm{Var}_{\boldsymbol{X}}[\hat{\lambda}^{\text{MME}}] = \lambda_0/n$.

[2] The class of the distribution is correctly-specified, but the true parameter value is unknown.

Sampling Distributions of MMEs

- The *bias* of an estimator $\hat{\theta}$ for θ is defined as $\mathrm{E}_{\mathbf{X}}[\hat{\theta}] - \theta$. In this example,

$$\mathrm{E}_{\mathbf{X}}[\hat{\lambda}^{\mathrm{MME}}] = \lambda_0. \tag{15}$$

  Estimators with zero bias are called *unbiased*.
- The *standard error* of an estimator $\hat{\theta}$ is defined as its standard deviation $\sqrt{\mathrm{Var}_{\mathbf{X}}[\hat{\theta}]} = \sigma_{\hat{\theta}}$. In this example,

$$\sigma_{\hat{\lambda}^{\mathrm{MME}}} = \sqrt{\lambda_0/n}. \tag{16}$$

- As long as we correctly specify the class of the underlying distribution, we may verify the unbiasedness of an estimator. In contrast, we cannot calculate the standard error without knowing λ_0.
- In such cases, we may use the *estimated standard error*:

$$s_{\hat{\lambda}^{\mathrm{MME}}} = \sqrt{\hat{\lambda}^{\mathrm{MME}}/n}. \tag{17}$$

Sampling Distributions of MMEs

Exercise 1

Example 8.4.B (Cont.): When $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_0, \sigma_0^2)$, we can use the MME $\hat{\sigma^2}^{\text{MME}} = \overline{X^2} - \left(\bar{X}\right)^2$ to estimate σ_0^2. Show that:

$$\frac{n\hat{\sigma^2}^{\text{MME}}}{\sigma_0^2} \sim \chi^2(n-1). \tag{18}$$

Sampling Distributions of MMEs

- In general, deriving exact sampling distributions is challenging. A common approach is to derive limiting distributions.
- We first define *consistency*, a desirable asymptotic property of estimators.
- **Consistency**: Let $(\hat{\theta}_n)_{n=1}^{\infty}$ be a sequence of estimators[3] for a parameter θ. Then, $\hat{\theta}_n$ is said to be *consistent* if $\hat{\theta}_n \xrightarrow{p} \theta$.
- For example, the sample mean $\bar{X} \xrightarrow{p} \mathrm{E}[X_1] = \mu$ by the Weak Law of Large Numbers. Thus, $\hat{\theta}_n = \bar{X}$ is a consistent estimator for $\theta = \mu$.
- Note that unbiasedness does not imply consistency. For example, the estimator $\hat{\theta}_n = X_1$ for μ is unbiased but inconsistent.

[3] Strictly speaking, an estimator is a rule that maps $(X_i)_{i=1}^{n}$ to a value for any sample size n (e.g., the sample mean maps $X_1 \to X_1$, $(X_1, X_2) \to (X_1 + X_2)/2$, ...), and the specific calculated value is the *estimate*. However, we refer to the sequence $\hat{\theta}_n$ as the estimator.

Sampling Distributions of MMEs

Exercise 2

Let X_i be i.i.d. samples with finite and positive variance σ^2. Show that the following MME is biased but consistent:

$$\hat{\sigma^2}^{\text{MME}} = \frac{n-1}{n} S^2 = \frac{1}{n} \sum_{i=1}^{n} \left(X_i - \bar{X} \right)^2 . \tag{19}$$

Sampling Distributions of MMEs

Exercise 3

Let $\theta = g(m_1, \ldots, m_r)$, where g is a continuous function, $r \in \mathbb{N}$, and the r-th population moment is finite (i.e., $m_r < \infty$). Show that the following MME is consistent:

$$\hat{\theta}^{\text{MME}} = g\left(\hat{m}_1, \ldots, \hat{m}_r\right). \tag{20}$$

Sampling Distributions of MMEs

- The notion of consistency can be extended to cases where the target quantity depends on n.
- **Example 8.4.A (Cont.)**: When $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Poi}(\lambda_0)$, we estimate the standard error $\sigma_{\hat{\lambda}^{\text{MME}}} = \sqrt{\lambda_0/n}$ using $s_{\hat{\lambda}^{\text{MME}}} = \sqrt{\hat{\lambda}^{\text{MME}}/n}$. Since $\hat{\lambda}^{\text{MME}}$ is consistent for λ_0, it follows that:

$$\frac{s_{\hat{\lambda}^{\text{MME}}}}{\sigma_{\hat{\lambda}^{\text{MME}}}} \xrightarrow{p} 1. \tag{21}$$

Sampling Distributions of MMEs

- While consistency guarantees the convergence of estimators to specific values, statistical inference often requires assessing the distributional behavior of estimators.
- For example, $\hat{m}_1$ is a consistent estimator for m_1. Furthermore, by the CLT, if $m_2 < \infty$, the following holds:

$$\sqrt{n}\left(\hat{m}_1 - m_1\right) \xrightarrow{d} \sqrt{m_2 - m_1^2}\, Z, \quad \text{where } Z \sim \mathcal{N}(0,1). \tag{22}$$

  Therefore, for large n,

$$P\left(|\hat{m}_1 - m_1| \leq z s_{\hat{m}_1}\right) \approx \Phi(z) - \Phi(-z). \tag{23}$$

- Similarly, if $m_{2r} < \infty$, we have:

$$\sqrt{n}\left(\hat{m}_r - m_r\right) \xrightarrow{d} \sqrt{m_{2r} - m_r^2}\, Z, \quad \text{where } Z \sim \mathcal{N}(0,1). \tag{24}$$

Sampling Distributions of MMEs

- When an MME is not in the form of a sample mean, the CLT is not directly applicable.
- However, since an MME is a function of sample moments (which are sample means of powers of the random variables), the following theorem is useful.
- **Delta Method**: Let $(X_n)_{n\geq 1}$ be a sequence of random variables and θ be a constant. Suppose that $\sqrt{n}\,(X_n - \theta) \xrightarrow{d} Z$ for some random variable Z. Then, for any continuously differentiable function g,

$$\sqrt{n}\,\big(g(X_n) - g(\theta)\big) \xrightarrow{d} g'(\theta)Z. \tag{25}$$

Sampling Distributions of MMEs

Exercise 4

Delta Method: Let $(X_n)_{n\geq 1}$ be a sequence of random variables and θ be a constant. Suppose that $\sqrt{n}\,(X_n - \theta) \xrightarrow{d} Z$ for some random variable Z. Show that, for any continuously differentiable function g,

$$\sqrt{n}\left(g(X_n) - g(\theta)\right) \xrightarrow{d} g'(\theta)Z. \tag{26}$$

Sampling Distributions of MMEs

Sampling Distributions of MMEs

Exercise 5

Example 8.4.A (Cont.): Let $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Poi}(\lambda_0)$. We estimate the standard error $\sigma_{\hat{\lambda}^{\text{MME}}} = \sqrt{\lambda_0/n}$ using $s_{\hat{\lambda}^{\text{MME}}} = \sqrt{\hat{\lambda}^{\text{MME}}/n}$, where $\hat{\lambda}^{\text{MME}} = \hat{m}_1$. Show that:

$$2n\left(s_{\hat{\lambda}^{\text{MME}}} - \sigma_{\hat{\lambda}^{\text{MME}}}\right) \xrightarrow{d} Z, \quad Z \sim \mathcal{N}(0,1). \tag{27}$$

Sampling Distributions of MMEs

- As we saw in Example 8.4.A, standard errors of estimators often depend on unknown parameters, so we cannot directly use them.
- For example, the convergence $\sqrt{n}\left(\hat{\lambda}^{\text{MME}} - \lambda_0\right) \xrightarrow{d} \sqrt{\lambda_0} Z$ (where $\hat{\lambda}^{\text{MME}} = \hat{m}_1$ and $Z \sim \mathcal{N}(0,1)$) implies that for a sufficiently large n:

$$P\left(|\hat{\lambda}^{\text{MME}} - \lambda_0| \leq z_{\alpha/2}\sigma_{\hat{\lambda}^{\text{MME}}}\right) \approx 1 - \alpha, \tag{28}$$

  where z_α represents the upper α-quantile of the standard normal distribution.
- Instead of estimating $\sigma_{\hat{\lambda}^{\text{MME}}}$, we can apply a *Variance Stabilizing Transformation* to obtain a transformed estimator whose approximate standard error is independent of unknown parameters.

Sampling Distributions of MMEs

- Let $g(\lambda) = 2\sqrt{\lambda}$. Applying the delta method yields:

$$\sqrt{n}\left(2\sqrt{\hat{\lambda}^{\text{MME}}} - 2\sqrt{\lambda_0}\right) \xrightarrow{d} \frac{1}{\sqrt{\lambda_0}}\left(\sqrt{\lambda_0}Z\right) = Z. \tag{29}$$

- Therefore, for a sufficiently large n:

$$P\left(\left|\sqrt{\hat{\lambda}^{\text{MME}}} - \sqrt{\lambda_0}\right| \leq \frac{z_{\alpha/2}}{2\sqrt{n}}\right) \approx 1 - \alpha. \tag{30}$$

- Unlike the previous case where the error bound depends on the estimate $\hat{\lambda}^{\text{MME}}$, we can specify the required sample size to bound the error. For example, if

$$n \geq \frac{z_{\alpha/2}^2}{4\epsilon^2}, \tag{31}$$

then the inequality $\left|\sqrt{\hat{\lambda}^{\text{MME}}} - \sqrt{\lambda_0}\right| \leq \epsilon$ holds with probability approximately $1 - \alpha$.

Sampling Distributions of MMEs

- The Delta method can be extended to multivariate random vectors. We omit the proof here.
- <u>**Delta Method for Random Vectors**</u>: Let $(\boldsymbol{X}_n)_{n\geq 1}$ be a sequence of random vectors and θ be a constant vector. Suppose that $\sqrt{n}\,(\boldsymbol{X}_n - \theta) \xrightarrow{d} \boldsymbol{Z}$ for some random vector $\boldsymbol{Z}$. Then, for any continuously differentiable function g:

$$\sqrt{n}\left(g(\boldsymbol{X}_n) - g(\theta)\right) \xrightarrow{d} \left(\nabla g(\theta)\right)^T \boldsymbol{Z}, \tag{32}$$

where $\nabla g(\theta)$ denotes the gradient vector of g at θ.

Sampling Distributions of MMEs

Exercise 6

Let $\theta = g(m_1, \ldots, m_r)$, where g is a continuously differentiable function, $r \in \mathbb{N}$, and the $(2r)$-th population moment is finite (i.e., $m_{2r} < \infty$). Let $\hat{\theta}^{\text{MME}} = g(\hat{m}_1, \ldots, \hat{m}_r)$ and $\boldsymbol{m} = (m_1, \ldots, m_r)^T$. Show that:

$$\sqrt{n}\left(\hat{\theta}^{\text{MME}} - \theta\right) \xrightarrow{d} \left(\nabla g(\boldsymbol{m})^T M \nabla g(\boldsymbol{m})\right)^{1/2} Z, \quad Z \sim \mathcal{N}(0,1) \tag{33}$$

where M is the covariance matrix with entries $(M)_{ij} = m_{i+j} - m_i m_j$.

Sampling Distributions of MMEs

Table of Contents

1. Introduction
2. Fitting the Poisson Distribution to Emissions of Alpha Particles
3. Parameter Estimation
4. The Method of Moments
5. The Method of Maximum Likelihood
6. The Bayesian Approach to Parameter Estimation
7. Efficiency and the Cramér-Rao Lower Bound
8. Sufficiency

The Method of Maximum Likelihood

- So far, we have derived various properties of MMEs, including consistency and asymptotic normality.
- Although MMEs possess many desirable properties, it remains unclear whether they are the optimal choice for estimating a given parameter. Furthermore, MMEs are not unique. There can be multiple MMEs for the same parameter.
- In this section, we introduce another fundamental approach: Maximum Likelihood Estimation.
- In Section 8.7 (Efficiency and the Cramér-Rao Lower Bound), we will demonstrate that maximum likelihood estimation provides optimal estimators under certain regularity conditions.

The Method of Maximum Likelihood

- Suppose $x_1, \ldots, x_n$ are i.i.d. observations from an unknown continuous distribution. By viewing them as a realization of $\boldsymbol{X} = (X_1, \ldots, X_n)^T$, we consider a parametric family of candidate PDFs:

$$\{f(\cdot;\theta) \mid \theta \in \Omega\} \tag{34}$$

  where θ and Ω represent the parameter and the parameter space, respectively.
- Parameter estimation is essentially the process of finding the most plausible PDF within this family.
- For a given θ, we evaluate the "plausibility" of the observed data using the joint PDF:

$$f_{\boldsymbol{X}}(\boldsymbol{x};\theta) = \prod_{i=1}^{n} f(x_i;\theta). \tag{35}$$

The Method of Maximum Likelihood

- For fixed data $\boldsymbol{x}$, the inequality $f_{\boldsymbol{X}}(\boldsymbol{x};\theta) > f_{\boldsymbol{X}}(\boldsymbol{x};\theta')$ indicates that the observed data is more "plausible" under θ than under θ'.
- In this context, the *likelihood function* is defined as:

$$L(\theta; x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i;\theta). \tag{36}$$

The Maximum Likelihood Estimate (MLE), denoted by $\hat{\theta}^{\text{MLE}}$, is defined as:

$$\hat{\theta}^{\text{MLE}} \in \underset{\theta\in\Omega}{\operatorname{argmax}}\, L(\theta; x_1, \ldots, x_n). \tag{37}$$

- The logarithm of the likelihood is referred to as the *log-likelihood*, denoted by:

$$l(\theta; x_1, \ldots, x_n) = \log L(\theta; x_1, \ldots, x_n) = \sum_{i=1}^{n} \log f(x_i;\theta). \tag{38}$$

In this class, log denotes the natural logarithm with base e. For brevity, we often write $l(\theta)$.

The Method of Maximum Likelihood

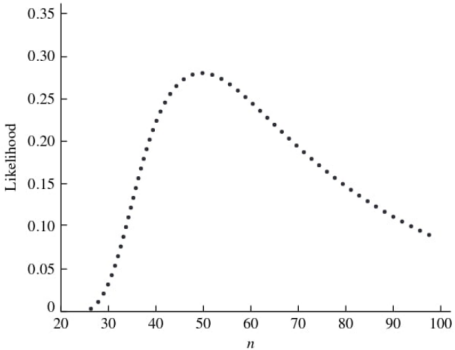


Fig. 1.3. [MSDA]: Likelihood Plot.

- **Revisiting Example 1.4.I**: Suppose that 10 animals are captured, tagged, and released. Later, 20 animals are captured, and $x = 4$ of them are found to be tagged.
- The probability of the observation x given a population size n, the likelihood $L(n; x)$, can be expressed as:

$$L(n; x) = \frac{\binom{10}{x}\binom{n-10}{20-x}}{\binom{n}{20}}. \tag{39}$$

- The likelihood $L(n; x = 4)$ is maximized at $n = 50$ (i.e., the MLE is $\hat{n}^{\text{MLE}} = 50$).

The Method of Maximum Likelihood

- **Example 8.5.A (Poisson Distribution)**: When $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Poi}(\lambda)$, the parametric family of model PMFs can be expressed as:

$$\left\{ p(x;\lambda) = \frac{\lambda^x e^{-\lambda}}{x!} \mid \lambda \in \Omega \right\}, \quad \text{where } \Omega = [0, \infty).^4 \tag{40}$$

- Given observed data $x_1, \ldots, x_n$, the log-likelihood is given by:

$$l(\lambda; x_1, \ldots, x_n) = \sum_{i=1}^{n} \log p(x_i;\lambda) = -\log \prod_{i=1}^{n} x_i! + n\bar{x} \log \lambda - n\lambda. \tag{41}$$

[4] Here, the parameter space $\Omega = [0, \infty)$ includes the degenerate case $\lambda = 0$.

The Method of Maximum Likelihood

- **Case 1 ($X_i > 0$ for some i)**: The derivative with respect to (w.r.t.) λ is:

$$l'(\lambda; x_1, \ldots, x_n) = \frac{n\bar{x}}{\lambda} - n = \frac{n(\bar{x} - \lambda)}{\lambda}. \tag{42}$$

  We observe that $l'(\lambda) > 0$ if and only if $0 < \lambda < \bar{x}$. Since $l(\lambda)$ increases on $(0, \bar{x})$ and decreases on $(\bar{x}, \infty)$, the global maximum occurs at $\hat{\lambda}^{\text{MLE}} = \bar{x}$.
- **Case 2 ($X_i = 0$ for all i)**: Since $\bar{x} = 0$, the log-likelihood simplifies to $l(\lambda) = -n\lambda$. This function is strictly decreasing on $[0, \infty)$, so it is maximized at the boundary $\hat{\lambda}^{\text{MLE}} = 0 = \bar{x}$.
- Thus, for the Poisson distribution, the MLE coincides with $\hat{\lambda}^{\text{MME}} = \bar{X}$.

The Method of Maximum Likelihood

Exercise 7

Example 8.5.B (Normal Distribution): Let $X_1, \ldots, X_n$ be i.i.d. random samples from $\mathcal{N}(\mu, \sigma^2)$. Find the MLE for $\theta = (\mu, \sigma^2)^T$.

The Method of Maximum Likelihood

- The following is a useful property for finding the MLE.
- Let I be an open interval. Suppose $l(\theta)$ is twice differentiable on I and $l''(\theta)$ is continuous on I. If $l''(\theta) < 0$ for all $\theta \in I$ and there exists $\hat{\theta} \in I$ such that $l'(\hat{\theta}) = 0$, then:

$$\hat{\theta} = \underset{\theta \in I}{\operatorname{argmax}}\, l(\theta). \tag{43}$$

- **Proof**: By Taylor's Theorem, for any $\theta \in I \setminus \{\hat{\theta}\}$, there exists $\tilde{\theta}$ between θ and $\hat{\theta}$ such that:

$$l(\theta) = l(\hat{\theta}) + l'(\hat{\theta})(\theta - \hat{\theta}) + \frac{1}{2} l''(\tilde{\theta})(\theta - \hat{\theta})^2. \tag{44}$$

Since $l'(\hat{\theta}) = 0$ and $l''(\tilde{\theta}) < 0$, we have:

$$l(\theta) = l(\hat{\theta}) + 0 + \frac{1}{2} l''(\tilde{\theta})(\theta - \hat{\theta})^2 < l(\hat{\theta}). \tag{45}$$

The Method of Maximum Likelihood

Exercise 8

Let $X_1, \ldots, X_n$ be i.i.d. samples from $\mathrm{Ber}(p)$. Find the MLE for $\theta = p$.

The Method of Maximum Likelihood

Exercise 9

Logistic Distribution with Location Parameter: Let $X_1, \ldots, X_n$ be i.i.d. random samples from a distribution with the following PDF:

$$f(x;\theta) = \frac{e^{x-\theta}}{(1+e^{x-\theta})^2}. \tag{46}$$

Show that the log-likelihood $l(\theta)$ is strictly concave and approaches $-\infty$ as $\theta \to -\infty$ or $\theta \to +\infty$.

The Method of Maximum Likelihood

The Method of Maximum Likelihood

- As seen in the previous exercise, MLEs often do not have closed-form expressions.
- In such cases, we may apply an iterative optimization method such as the Newton-Raphson method:

$$\hat{\theta}^{(t+1)} = \hat{\theta}^{(t)} + \left[-l''(\hat{\theta}^{(t)})\right]^{-1} l'(\hat{\theta}^{(t)}), \quad t = 0, 1, \ldots \tag{47}$$

- A typical choice for the initial value $\hat{\theta}^{(0)}$ is the MME. For example, in the previous exercise, since $\mathrm{E}[X] = \theta$, we may use $\hat{\theta}^{(0)} = \hat{\theta}^{\mathrm{MME}} = \hat{m}_1$.

The Method of Maximum Likelihood

- Let $\hat{\theta}^{\text{MLE}}$ exist. Then, the MLE for $\eta = g(\theta)$, where g is a one-to-one function, is given by:

$$\hat{\eta}^{\text{MLE}} = g(\hat{\theta}^{\text{MLE}}). \tag{48}$$

- For $\mathcal{N}(\mu, \sigma^2)$, we derived that the MLE for $\theta = (\mu, \sigma^2)^T$ is:

$$\hat{\theta}^{\text{MLE}} = \left(\bar{X}, n^{-1} \sum_{i=1}^{n} (X_i - \bar{X})^2 \right)^T. \tag{49}$$

With the transformation $g(\mu, \sigma^2) = (\mu, \sigma)^T$, the MLE for $\eta = (\mu, \sigma)^T$ is given by:

$$\hat{\eta}^{\text{MLE}} = \left(\bar{X}, \sqrt{n^{-1} \sum_{i=1}^{n} (X_i - \bar{X})^2} \right)^T. \tag{50}$$

Exponential Family

- Now, we define a broad family of parametric distributions to study a unified approach to finding the MLE.
- As motivating examples, the PMF of $\mathrm{Ber}(p)$ and the PDF of $\mathrm{Exp}(\lambda)$ can be expressed as:
  - $\mathrm{Ber}(p)$: $p(x;p) = p^x(1-p)^{1-x} = \exp\left(\left(\log \frac{p}{1-p}\right)x + \log(1-p)\right), \quad x \in \{0,1\}.$
  - $\mathrm{Exp}(\lambda)$: $f(x;\lambda) = \lambda e^{-\lambda x} = \exp\left(-\lambda x + \log \lambda\right), \quad x \in \{0,1,\dots\}.$
- Note that the exponent is a linear function of x, where the coefficient of x is $\eta(p) = \log \frac{p}{1-p}$ or $\eta(\lambda) = -\lambda$. Here, η is called the *natural parameter*, which directly governs the shape of the distribution.

Exponential Family

- **Single Parameter Exponential Family**: A random variable X follows a *single parameter exponential family distribution* if its PDF (or PMF) is of the form:

$$f_X(x;\eta) = h(x)\exp\left(\eta T(x) - A(\eta)\right) \tag{51}$$

where $\eta = \eta(\theta)$ is the *natural parameter*, $T(x)$ is a statistic, and $A(\eta)$ is the log-partition function, satisfying the following regularity conditions:

(C1) The natural parameter space $\eta(\Omega)$ is an open interval.
(C2) $T(X)$ is not degenerate. That is, $\mathrm{Var}_\eta[T(X)] > 0$ for all $\eta \in \eta(\Omega)$.

- We denote the expectation and variance w.r.t. $f_X(x;\eta)$ as $\mathrm{E}_\eta[T(X)] = \int_{-\infty}^{\infty} T(x) f_X(x;\eta)dx$ and $\mathrm{Var}_\eta[T(X)] = \int_{-\infty}^{\infty}(T(x) - \mathrm{E}_\eta[T(X)])^2 f_X(x;\eta)dx$, respectively.

Exponential Family

- When $f_X(x;\eta)$ (or $p_X(x;\eta)$) is given by Equation (51), the support $\mathcal{X} = \{x : f_X(x;\eta) > 0\}$ is determined solely by $h(x)$, and thus does not depend on η.
- Examples of single parameter exponential family distributions satisfying (C1) and (C2) include:
  - $\mathrm{Ber}(p)$ where $\theta = p$
  - $\mathrm{Bin}(n, p)$ where $\theta = p$ (assuming n is known)
  - $\mathrm{Poi}(\lambda)$ where $\theta = \lambda$
  - $\mathrm{Exp}(\lambda)$ where $\theta = \lambda$

Exponential Family

Exercise 10

Let X follow an exponential family distribution whose PDF is given by:

$$f_X(x;\eta) = h(x)\exp\left(\eta T(x) - A(\eta)\right), \tag{52}$$

satisfying the following regularity conditions:

(C1) The natural parameter space $\eta(\Omega)$ is an open interval.

(C2) $T(X)$ is not degenerate. That is, $\mathrm{Var}_\eta[T(X)] > 0$ for all $\eta \in \eta(\Omega)$.

Exponential Family

a) Show that $A'(\eta) = \mathrm{E}_\eta[T(X)]$ and $A''(\eta) = \mathrm{Var}_\eta[T(X)]$.

Exponential Family

b) Show that $\hat{\eta} \in \eta(\Omega)$ is the unique MLE if it satisfies the following *likelihood equation*:

$$\mathrm{E}_{\hat{\eta}}[T(X)] = \overline{T(X)} := n^{-1} \sum_{i=1}^{n} T(X_i). \tag{53}$$

Exponential Family: Bernoulli

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{Ber}(p)$, where $\theta = p$ and $\Omega = [0, 1]$. Its PMF is given by:

$$p_X(x; p) = p^x (1-p)^{1-x} = \exp\left(\left(\log \frac{p}{1-p}\right) x + \log(1-p)\right), \quad x \in \{0, 1\}, \tag{54}$$

for $\theta \in \Omega_o = (0, 1)$. Let $\eta(\theta) = \log \frac{p}{1-p}$. Then, the PMF can be expressed in the exponential family form:

$$p_X(x; \eta) = \exp\left(\eta x - \log(1 + \exp(\eta))\right), \quad x \in \{0, 1\}. \tag{55}$$

Here,

$$h(x) = \begin{cases} 1, & \text{if } x \in \{0, 1\}, \\ 0, & \text{otherwise}, \end{cases} \tag{56}$$

$T(x) = x$, and $A(\eta) = \log(1 + \exp(\eta))$.

Exponential Family: Bernoulli

- Since $\eta(\Omega_o) = (-\infty, \infty)$ and $\mathrm{Var}_\eta[T(X)] = \frac{e^\eta}{(1+e^\eta)^2} > 0$, if there exists $\hat{\eta}$ satisfying the likelihood equation:

$$\mathrm{E}_{\hat{\eta}}[T(X)] = \overline{T(X)} \iff \frac{e^{\hat{\eta}}}{1+e^{\hat{\eta}}} = \bar{X}, \tag{57}$$

  then $\hat{\eta}$ is the unique MLE over $\eta(\Omega_o)$.
- **Case 1** ($0 < \bar{X} < 1$): The solution is $\hat{\eta}^{\mathrm{MLE}} = \log\left(\frac{\bar{X}}{1-\bar{X}}\right)$. Since the mapping $\eta \to p$ is one-to-one and the likelihood $L(p)$ is continuous on $[0, 1]$, we have $\hat{p}^{\mathrm{MLE}} = \frac{e^{\hat{\eta}^{\mathrm{MLE}}}}{1+e^{\hat{\eta}^{\mathrm{MLE}}}} = \bar{X}$.
- **Case 2** ($\bar{X} = 0$): There is no solution for $\hat{\eta}$ in the open interval $\eta(\Omega_o)$. However, direct maximization of $l(p) = n\log(1-p)$ yields $\hat{p}^{\mathrm{MLE}} = 0$.
- **Case 3** ($\bar{X} = 1$): There is no solution for $\hat{\eta}$ in the open interval $\eta(\Omega_o)$. However, direct maximization of $l(p) = n\log p$ yields $\hat{p}^{\mathrm{MLE}} = 1$.
- Thus, in all cases, $\hat{p}^{\mathrm{MLE}} = \bar{X}$.

Exponential Family: Poisson

- Let $X_1, \ldots, X_n$ be i.i.d. random samples from $\mathrm{Poi}(\lambda)$, where $\theta = \lambda$ and $\Omega = [0, \infty)$. Its PMF is given by:

$$p_X(x; \lambda) = \frac{\lambda^x}{x!} e^{-\lambda} = \frac{1}{x!} \exp\left((\log \lambda)x - \lambda\right), \quad x \in \{0, 1, \ldots\}, \tag{58}$$

for $\theta \in \Omega_o = (0, \infty)$. Let $\eta(\theta) = \log \lambda$. Then, the PMF can be expressed in the exponential family form:

$$p_X(x; \eta) = \frac{1}{x!} \exp\left(\eta x - e^{\eta}\right), \quad x \in \{0, 1, \ldots\}. \tag{59}$$

Here,

$$h(x) = \begin{cases} \frac{1}{x!}, & \text{if } x \in \{0, 1, \ldots\}, \\ 0, & \text{otherwise}, \end{cases} \tag{60}$$

$T(x) = x$, and $A(\eta) = e^{\eta}$.

Exponential Family: Poisson

- Since $\eta(\Omega_o) = (-\infty, \infty)$ and $\mathrm{Var}_\eta[T(X)] = e^\eta > 0$, if there exists $\hat{\eta}$ satisfying the likelihood equation:

$$\mathrm{E}_{\hat{\eta}}[T(X)] = \overline{T(X)} \iff e^{\hat{\eta}} = \bar{X}, \tag{61}$$

  then $\hat{\eta}$ is the unique MLE over $\eta(\Omega_o)$.
- **Case 1 ($\bar{X} > 0$)**: The solution is $\hat{\eta}^{\mathrm{MLE}} = \log \overline{X}$. Since the mapping $\eta \to \lambda$ is one-to-one and the likelihood $L(\lambda)$ is continuous at 0, we have $\hat{\lambda}^{\mathrm{MLE}} = \exp(\hat{\eta}^{\mathrm{MLE}}) = \bar{X}$.
- **Case 2 ($\bar{X} = 0$)**: There is no solution for $\hat{\eta}$ in the open interval $\eta(\Omega_o)$. However, direct maximization of $l(\lambda) = -n\lambda$ yields $\hat{\lambda}^{\mathrm{MLE}} = 0$.
- Thus, in both cases, $\hat{\lambda}^{\mathrm{MLE}} = \bar{X}$.

Exponential Family: Exponential

Exercise 11

Let $X_1, \ldots, X_n$ be i.i.d. random samples from $\mathrm{Exp}(\lambda)$, where $\theta = \lambda$ and $\Omega = (0, \infty)$. Find the MLE for θ.

Multi-parameter Exponential Family

- **Multi-parameter Exponential Family**: A random vector $\boldsymbol{X}$ follows a *multi-parameter exponential family distribution* if its PDF (or PMF) is of the form:

$$f_{\boldsymbol{X}}(\boldsymbol{x};\boldsymbol{\eta}) = h(\boldsymbol{x}) \exp\left(\boldsymbol{\eta}^\top \boldsymbol{T}(\boldsymbol{x}) - A(\boldsymbol{\eta})\right) \tag{62}$$

where $\boldsymbol{\eta} = \boldsymbol{\eta}(\theta)$ is the vector of natural parameters, $\boldsymbol{T}(\boldsymbol{x})$ is the vector of statistics, and $A(\boldsymbol{\eta})$ is the log-partition function. We assume the following regularity conditions:

(C1) The natural parameter space $\boldsymbol{\eta}(\Omega)$ is an open set in $\mathbb{R}^k$.

(C2) The covariance matrix $\mathrm{Var}_{\boldsymbol{\eta}}[\boldsymbol{T}(\boldsymbol{X})]$ is positive definite (non-singular) for all $\boldsymbol{\eta} \in \boldsymbol{\eta}(\Omega)$.

Multi-parameter Exponential Family

- Similar to the single parameter case, under conditions (C1) and (C2):

$$\nabla A(\boldsymbol{\eta}) = \mathrm{E}_{\boldsymbol{\eta}}[\boldsymbol{T}(X)] \quad \text{and} \quad \nabla^2 A(\boldsymbol{\eta}) = \mathrm{Var}_{\boldsymbol{\eta}}[\boldsymbol{T}(X)].$$

- The MLE $\hat{\boldsymbol{\eta}}^{\mathrm{MLE}}$ is the unique solution to the following likelihood equation (provided a solution exists within $\boldsymbol{\eta}(\Omega)$):

$$\mathrm{E}_{\hat{\boldsymbol{\eta}}}[\boldsymbol{T}(X)] = \overline{\boldsymbol{T}(X)} := n^{-1} \sum_{i=1}^{n} \boldsymbol{T}(X_i). \tag{63}$$

Furthermore, when $\boldsymbol{\eta}$ is one-to-one, the MLE $\hat{\theta}^{\mathrm{MLE}}$ is given by the unique solution of:

$$\mathrm{E}_{\hat{\theta}}[\boldsymbol{T}(X)] = \overline{\boldsymbol{T}(X)} := n^{-1} \sum_{i=1}^{n} \boldsymbol{T}(X_i). \tag{64}$$

Multi-parameter Exponential Family: Univariate Gaussian

Exercise 12

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, where $\theta = (\mu, \sigma^2)^\top$ and $\Omega = (-\infty, \infty) \times (0, \infty)$.

a) Show that its PDF belongs to the multi-parameter exponential family satisfying conditions (C1) and (C2).

Multi-parameter Exponential Family: Univariate Gaussian

b) Use the likelihood equation to find the MLE for θ.

Sampling Distributions of MLEs

- When MLEs have a closed-form expression, they often coincide with MMEs. Thus, we can directly apply the results on the sampling distributions of MMEs.
- In contrast, when the MLE lacks a closed-form expression, it is challenging to derive its exact sampling distribution.
- Nevertheless, like MMEs, MLEs exhibit consistency and asymptotic normality under regularity conditions. Furthermore, under these conditions, MLEs are asymptotically efficient (i.e., they have the smallest variance among consistent estimators).

Consistency of MLEs

- Here, we focus on continuous distributions.
- We first introduce the Kullback-Leibler (KL) divergence to prove consistency.
- **Kullback-Leibler Divergence**: Let $f_1(x)$ and $f_2(x)$ be PDFs such that $\text{supp}(f_1) \subseteq \text{supp}(f_2)$. The KL-divergence of f_2 from f_1 is defined as:

$$\mathcal{D}_{\text{KL}}(f_1 || f_2) = \int_{-\infty}^{\infty} f_1(x) \log \left(\frac{f_1(x)}{f_2(x)} \right) dx. \tag{65}$$

- Note that $g(t) = t \log t$ is a strictly convex function with $g(1) = 0$. The KL-divergence corresponds to the *f-divergence* generated by this function g.

Consistency of MLEs

Exercise 13

Show that $\mathcal{D}_{\mathrm{KL}}(f_1||f_2) \geq 0$, and that the equality holds if and only if $f_1 = f_2$.

Consistency of MLEs

- The KL divergence between $f(\cdot;\theta_0)$ and $f(\cdot;\theta)$ is given by:

$$\mathcal{D}_{\mathrm{KL}}(f(\cdot;\theta_0)||f(\cdot;\theta)) = \mathrm{E}_{\theta_0}\left[\log\frac{f(X;\theta_0)}{f(X;\theta)}\right] = \int_{-\infty}^{\infty}\left(\log\frac{f(x;\theta_0)}{f(x;\theta)}\right)f(x;\theta_0)dx.$$

- Note that the KL divergence measures the discrepancy between the distributions corresponding to θ_0 and θ.
- From the previous result, $\mathcal{D}_{\mathrm{KL}}(f(\cdot;\theta_0)||f(\cdot;\theta)) \geq 0$, with equality holding if and only if $f(\cdot;\theta_0) = f(\cdot;\theta)$. When PDFs identify their parameters, the equality holds if and only if $\theta = \theta_0$.

Consistency of MLEs

- **Consistency of MLEs**: Let $X_1, \ldots, X_n$ be i.i.d. random variables with PDF $f(x;\theta_0)$ where $\theta_0 \in \Omega$. Denoting $\bar{l}_n(\theta) = n^{-1}\sum_{i=1}^{n} \log f(X_i;\theta)$ and $\bar{l}_\infty(\theta) = \mathrm{E}_{\theta_0}[\log f(X;\theta)]$, we assume the following regularity conditions:
  - (R1) (Identifiability) For any $\theta \in \Omega$, if $f(\cdot;\theta) = f(\cdot;\theta_0)$, then $\theta = \theta_0$.
  - (R2) (Continuity) $\log f(x;\theta)$ is continuous with respect to θ for all x.
  - (R3) (Compactness) The parameter space Ω is compact.
  - (R4) (Uniform Convergence) $\sup_{\theta\in\Omega}|\bar{l}_n(\theta) - \bar{l}_\infty(\theta)| \xrightarrow{p} 0$. That is, for any $\epsilon > 0$,

$$\lim_{n\to\infty} P_{\theta_0}\left(\sup_{\theta\in\Omega}|\bar{l}_n(\theta) - \bar{l}_\infty(\theta)| \geq \epsilon\right) = 0. \tag{66}$$

Then, $\hat{\theta}_n^{\mathrm{MLE}} \xrightarrow{p} \theta_0$.[5]

[5] Note that for a given sample, the MLE may not be unique. This result implies that any sequence of choices for $\hat{\theta}_n^{\mathrm{MLE}}$ converges to θ_0 in probability.

Consistency of MLEs

- The key idea of the proof is as follows.
- First, the Weak Law of Large Numbers implies that for a given θ:

$$\bar{l}_n(\theta) \xrightarrow{p} \bar{l}_\infty(\theta). \tag{67}$$

  Condition (R4) provides the uniform version of this convergence over Ω.
- Next, we show that θ_0 is the unique maximizer of the limit function $\bar{l}_\infty(\theta)$. Note that $\hat{\theta}_n^{\text{MLE}}$ is defined as the maximizer of $\bar{l}_n(\theta)$.
- Finally, under the aforementioned regularity conditions, we demonstrate that the convergence of functions $\bar{l}_n$ to $\bar{l}_\infty$ implies the convergence of their maximizers, i.e., $\hat{\theta}_n^{\text{MLE}}$ converges to θ_0, the unique maximizer of $\bar{l}_\infty(\theta)$.

Consistency of MLEs

- **Remarks on Regularity Conditions**:
  - Condition (R2) implies that the support $\{x : f(x;\theta) > 0\}$ does not depend on θ.
  - Condition (R3) is often violated (e.g., $\Omega = [0, \infty)$ for Poisson). In such cases, we can construct a sequence of monotonically increasing compact subsets that converge to Ω.
  - Note that (R2) ensures the continuity of the log-likelihood function. Thus, combined with the compactness of Ω in (R3), the existence of $\hat{\theta}_n^{\text{MLE}}$ is guaranteed.[6]
  - Verifying (R4) directly is usually difficult. Instead, we usually assume a stronger condition called the *dominance condition*, which implies (R4): There exists a function $D(x)$ such that $|\log f(x;\theta)| \leq D(x)$ for all $\theta \in \Omega$ and $\mathrm{E}_{\theta_0}[D(X)] < \infty$.

[6] The Extreme Value Theorem states that a continuous real-valued function on a compact set attains its maximum and minimum values.

Consistency of MLEs

Exercise 14

Consistency of MLEs: Let $X_1, \ldots, X_n$ be i.i.d. random variables with PDF $f(x; \theta_0)$ where $\theta_0 \in \Omega$. Denoting $\bar{l}_n(\theta) = n^{-1} \sum_{i=1}^{n} \log f(X_i; \theta)$ and $\bar{l}_\infty(\theta) = \mathrm{E}_{\theta_0}[\log f(X; \theta)]$, we assume all the aforementioned regularity conditions.

a) Show that θ_0 is the unique maximizer of $\bar{l}_\infty(\theta)$.

Consistency of MLEs

b) Show that, for any $\epsilon > 0$, $\sup_{\theta \in \Omega : |\theta - \theta_0| \geq \epsilon} \bar{l}_\infty(\theta) < \bar{l}_\infty(\theta_0)$.

Consistency of MLEs

c) Let $\delta(\epsilon) = \bar{l}_\infty(\theta_0) - \sup_{\theta\in\Omega:|\theta-\theta_0|\geq\epsilon} \bar{l}_\infty(\theta)$. Show that if $\sup_{\theta\in\Omega}|\bar{l}_n(\theta) - \bar{l}_\infty(\theta)| < \delta(\epsilon)/2$, then $\bar{l}_\infty(\hat{\theta}_n^{\text{MLE}}) > \sup_{\theta\in\Omega:|\theta-\theta_0|\geq\epsilon} \bar{l}_\infty(\theta)$.

Consistency of MLEs

d) Show that, for any $\epsilon > 0$,

$$P_{\theta_0}\left(|\hat{\theta}_n^{\text{MLE}} - \theta_0| \geq \epsilon\right) \leq P_{\theta_0}\left(\sup_{\theta \in \Omega} |\bar{l}_n(\theta) - \bar{l}_\infty(\theta)| \geq \delta(\epsilon)/2\right). \tag{68}$$

Asymptotic Normality of MLEs

- With additional regularity conditions, we can derive the asymptotic normality of MLEs. This course provides a sketch of the proof rather than a rigorous derivation.
- Denoting $\bar{l}_n(\theta) = n^{-1}\sum_{i=1}^{n} \log f(X_i;\theta)$ and $\bar{l}_\infty(\theta) = \mathrm{E}_{\theta_0}[\log f(X;\theta)]$, the key idea is based on the Taylor expansion of $\bar{l}_n'(\hat{\theta}_n^{\mathrm{MLE}})$ around θ_0:

$$\bar{l}_n'(\hat{\theta}_n^{\mathrm{MLE}}) = \bar{l}_n'(\theta_0) + \bar{l}_n''(\theta_0)(\hat{\theta}_n^{\mathrm{MLE}} - \theta_0) + \frac{\bar{l}_n^{(3)}(\tilde{\theta}_n)}{2}\left(\hat{\theta}_n^{\mathrm{MLE}} - \theta_0\right)^2, \tag{69}$$

  where $\tilde{\theta}_n$ lies between θ_0 and $\hat{\theta}_n^{\mathrm{MLE}}$.
- Under regularity conditions, $\bar{l}_n'(\hat{\theta}_n^{\mathrm{MLE}}) = 0$ with high probability,[7] which implies:

$$\sqrt{n}(\hat{\theta}_n^{\mathrm{MLE}} - \theta_0) = -\left(\bar{l}_n''(\theta_0) + \frac{\bar{l}_n^{(3)}(\tilde{\theta}_n)}{2}(\hat{\theta}_n^{\mathrm{MLE}} - \theta_0)\right)^{-1}\sqrt{n}\bar{l}_n'(\theta_0). \tag{70}$$

[7] This holds with probability approaching 1 for any $\epsilon > 0$ such that $\{\theta : |\theta - \theta_0| < \epsilon\} \subseteq \Omega$.

Asymptotic Normality of MLEs: Fisher Information

- Therefore, the essential components in deriving asymptotic normality include the score function and Fisher Information.
- **Score Function**: The *score function* $s_n(\theta)$ is defined as the gradient of the log-likelihood with respect to θ:

$$s_n(\theta) = \sum_{i=1}^{n} \frac{\partial}{\partial\theta} \log f(X_i;\theta) = n\bar{l}_n'(\theta). \tag{71}$$

- **Fisher Information**: The *Fisher Information* $\mathcal{I}_n(\theta)$ is defined as the variance of the score function:

$$\mathcal{I}_n(\theta) = \mathrm{Var}_\theta[s_n(\theta)]. \tag{72}$$

- Since the observations X_i are i.i.d., the Fisher Information is additive:

$$\mathcal{I}_n(\theta) = n\mathcal{I}_1(\theta) = n\,\mathrm{Var}_\theta\left[\frac{\partial}{\partial\theta}\log f(X;\theta)\right] = n\,\mathrm{Var}_\theta[\bar{l}_1'(\theta)]. \tag{73}$$

Asymptotic Normality of MLEs: Fisher Information

Exercise 15

Let X be a random variable with PDF $f(x;\theta)$. We assume the following regularity conditions hold:

(R2)' (Smoothness) $\log f(x;\theta)$ is **three times continuously differentiable** w.r.t. θ for all x.

(R5) (Fisher Information) For any $\theta \in \Omega$, $\mathcal{I}_1(\theta)$ is finite and positive:

$$0 < \mathcal{I}_1(\theta) = \mathrm{Var}_\theta\left[\frac{\partial}{\partial\theta}\log f(X;\theta)\right] < \infty.$$

Asymptotic Normality of MLEs: Fisher Information

a) Show that

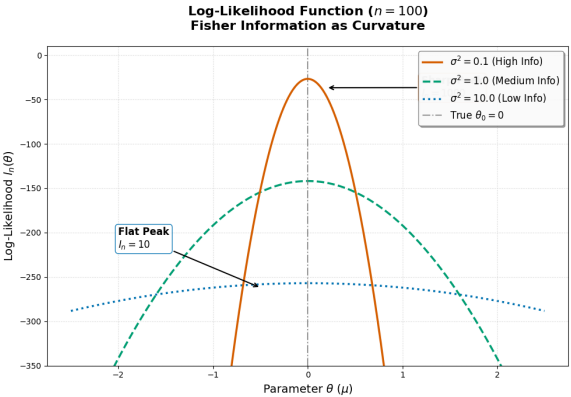
$$\mathrm{E}_\theta\left[\frac{\partial}{\partial\theta}\log f(X;\theta)\right] = 0. \tag{74}$$

Asymptotic Normality of MLEs: Fisher Information

b) Show that

$$\mathcal{I}_1(\theta) = \mathrm{E}_\theta\left[-\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\right]. \tag{75}$$

Asymptotic Normality of MLEs: Fisher Information



- $\mathcal{I}_1(\theta) = -\mathrm{E}_\theta[l_1''(\theta)]$ represents the expected curvature of $l_n(\theta)$.
- For $\mathcal{N}(\mu, \sigma^2)$ where $\theta = \mu$ (estimating μ with fixed σ^2), larger $\mathcal{I}_1(\theta) = \frac{1}{\sigma^2}$ indicates that $l_n(\theta)$ has a sharper peak.

Asymptotic Normality of MLEs: Regularity Conditions

- **Asymptotic Normality of MLEs**: Let $X_1, \ldots, X_n$ be i.i.d. random variables with PDF $f(x;\theta_0)$ where $\theta_0 \in$ **int(Ω)**. We assume:

(R1,3,4) (R1: Identifiability), (R3: Compactness), and (R4: Uniform Convergence) hold.

(R2)' (Smoothness) $\log f(x;\theta)$ is **three times continuously differentiable** w.r.t. θ for all x.

(R5) (Fisher Information) For any $\theta \in \Omega$, $\mathcal{I}_1(\theta)$ is finite and positive:

$$0 < \mathcal{I}_1(\theta) = \mathrm{Var}_\theta\left[\frac{\partial}{\partial\theta}\log f(X;\theta)\right] < \infty.$$

(R6) (Boundedness) There exists a function $H(x)$ such that $\sup_{\theta\in\Omega}\left|\frac{\partial^3}{\partial\theta^3}\log f(x;\theta)\right| \le H(x)$, with $\mathrm{E}_{\theta_0}[H(X)] < \infty$.

Then, $\sqrt{n}(\hat{\theta}_n^{\mathrm{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, \mathcal{I}_1(\theta_0)^{-1}\right)$.

Asymptotic Normality of MLEs: A Sketch of the Proof

- The consistency of MLEs, together with the additional regularity conditions, implies:

$$\sqrt{n}(\hat{\theta}_n^{\text{MLE}} - \theta_0) = -\Bigg(\underbrace{\bar{l}_n''(\theta_0) + \frac{\bar{l}_n^{(3)}(\tilde{\theta}_n)}{2}(\hat{\theta}_n^{\text{MLE}} - \theta_0)}_{\xrightarrow{p} -\mathcal{I}_1(\theta_0)}\Bigg)^{-1} \underbrace{\sqrt{n}\bar{l}_n'(\theta_0)}_{\xrightarrow{d} \mathcal{N}(0, \mathcal{I}_1(\theta_0))} \tag{76}$$

$$\xrightarrow{d} \mathcal{N}\left(0, \frac{1}{(\mathcal{I}_1(\theta_0))^2} \times \mathcal{I}_1(\theta_0)\right) = \mathcal{N}\left(0, \mathcal{I}_1(\theta_0)^{-1}\right).$$

Asymptotic Normality of MLEs: A Sketch of the Proof

Exercise 16

Asymptotic Normality of MLEs: Let $X_1, \ldots, X_n$ be i.i.d. random variables with PDF $f(x; \theta_0)$ where $\theta_0 \in \mathbf{int(\Omega)}$. We assume all the aforementioned regularity conditions hold. Show that:

$$\bar{l}_n''(\theta_0) + \frac{\bar{l}_n^{(3)}(\tilde{\theta}_n)}{2}(\hat{\theta}_n^{\text{MLE}} - \theta_0) \xrightarrow{p} -\mathcal{I}_1(\theta_0). \tag{77}$$

Asymptotic Normality of MLEs: A Sketch of the Proof

Remark on MLEs

- The asymptotic normality implies that we can construct an approximate $(1-\alpha)$ confidence interval:

$$P_{\theta_0}\left(|\hat{\theta}_n^{\text{MLE}} - \theta_0| \leq \frac{z_{\alpha/2}}{\sqrt{\mathcal{I}_n(\theta_0)}}\right) \approx 1-\alpha. \tag{78}$$

- Recall that Fisher Information $\mathcal{I}_n(\theta_0)$ measures the curvature of the log-density.
- If the log-density is highly curved (sharp peak) at θ_0, $\mathcal{I}_n(\theta_0)$ is large. Consequently, the asymptotic variance $(\mathcal{I}_n(\theta_0))^{-1}$ is small, leading to a narrower confidence interval (i.e., more precise estimation).
- In Section 8.7, we will demonstrate that under certain conditions, no other consistent estimator (including MMEs) yields a narrower confidence interval than the MLE.

Remark on MLEs

- So far, we have derived various asymptotic properties of MLEs under regularity conditions.
- Even when the model is misspecified or some of the regularity conditions are violated, the method of maximum likelihood can still be justified as minimizing the KL-divergence.
- Specifically, let f_0 be the true PDF such that $\text{supp}(f_0) \subseteq \text{supp}(f(\cdot;\theta))$. Then:

$$\begin{aligned} \mathcal{D}_{\mathrm{KL}}(f_0 || f(\cdot;\theta)) &= \int_{-\infty}^{\infty} \left(\log \frac{f_0(x)}{f(x;\theta)} \right) f_0(x) dx \\ &= -\mathrm{E}_0[\log f(X;\theta)] + \mathrm{E}_0[\log f_0(X)] \\ &\approx -\bar{l}_n(\theta) + \text{Constant}. \end{aligned} \tag{79}$$

- Thus, maximizing $\bar{l}_n(\theta)$ is equivalent to minimizing the empirical KL-divergence.

Table of Contents

1. Introduction
2. Fitting the Poisson Distribution to Emissions of Alpha Particles
3. Parameter Estimation
4. The Method of Moments
5. The Method of Maximum Likelihood
6. The Bayesian Approach to Parameter Estimation
7. Efficiency and the Cramér-Rao Lower Bound
8. Sufficiency

The Bayesian Approach to Parameter Estimation

- Until now, we have viewed MMEs and MLEs as solutions of optimization problems that search the parameter space Ω.
- In contrast, the Bayesian approach views θ as a realization of a random variable Θ and derives the conditional distribution of Θ given $X_1, \ldots, X_n$ to model its stochastic uncertainty.
- Specifically, with the *prior distribution* $f_\Theta(\theta)$ and the *likelihood* $f_{X_1,\ldots,X_n|\Theta}(x_1, \ldots, x_n \mid \theta)$, the *posterior distribution* $f_{\Theta|X_1,\ldots,X_n}(\theta \mid x_1, \ldots, x_n)$ can be expressed as:

$$\begin{aligned} f_{\Theta|X_1,\ldots,X_n}(\theta \mid x_1, \ldots, x_n) &= C(x_1, \ldots, x_n) f_{X_1,\ldots,X_n|\Theta}(x_1, \ldots, x_n \mid \theta) f_\Theta(\theta) \\ &\propto f_{X_1,\ldots,X_n|\Theta}(x_1, \ldots, x_n \mid \theta) f_\Theta(\theta). \end{aligned} \tag{80}$$

The Bayesian Approach to Parameter Estimation

- **Example 8.6.A**: Let $\Lambda \sim \mathrm{Gamma}(\alpha, v)$ and $X_i \mid \Lambda = \lambda \overset{\text{i.i.d.}}{\sim} \mathrm{Poi}(\lambda)$. Then,

$$\begin{aligned} f_{\Lambda|X_1,\ldots,X_n}(\lambda \mid x_1,\ldots,x_n) &\propto f_{X_1,\ldots,X_n|\Lambda}(x_1,\ldots,x_n \mid \lambda) f_\Lambda(\lambda) \\ &\propto \left(\lambda^{\sum_{i=1}^n x_i} e^{-n\lambda}\right)\left(\lambda^{\alpha-1} e^{-v\lambda}\right) \\ &= \lambda^{(n\bar{x}+\alpha)-1} e^{-(n+v)\lambda}. \end{aligned} \tag{81}$$

  Therefore, $\Lambda \mid X_1,\ldots,X_n \sim \mathrm{Gamma}\left(n\bar{X}+\alpha, n+v\right)$.

- The *posterior mean* and *posterior variance* are given by:

$$\mathrm{E}_{\Lambda|X_1,\ldots,X_n}[\Lambda \mid X_1,\ldots,X_n] = \frac{n\bar{X}+\alpha}{n+v} \tag{82}$$

  and

$$\mathrm{Var}_{\Lambda|X_1,\ldots,X_n}[\Lambda \mid X_1,\ldots,X_n] = \frac{n\bar{X}+\alpha}{(n+v)^2}. \tag{83}$$

The Bayesian Approach to Parameter Estimation

- When using the posterior mean as an estimate for λ, we have:

$$\hat{\lambda}^{\text{Bayes}} = \underbrace{\left(\frac{n}{n+v}\right)}_{\text{Weight on MLE}} \bar{X} + \underbrace{\left(\frac{v}{n+v}\right)}_{\text{Weight on Prior}} \frac{\alpha}{v}. \tag{84}$$

- $\hat{\lambda}^{\text{Bayes}}$ is a weighted average of $\bar{X}$ and $\alpha/v = \mathrm{E}_{\Lambda}[\Lambda]$.
- As $n \to \infty$, the influence of the prior vanishes, and the estimator is approximately the MLE. Furthermore, the posterior variance converges to 0, implying that uncertainty in the posterior distribution vanishes.

The Bayesian Approach to Parameter Estimation

Exercise 17

Example 8.6.B: Let $\Theta \sim \mathcal{N}(\theta_0, \tau_{\text{prior}}^{-1})$ and $X_i \mid \Theta = \theta \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, \tau_0^{-1})$ for some constants $\theta_0 \in \mathbb{R}$, $\tau_{\text{prior}} > 0$, and $\tau_0 > 0$. Show that:

$$\Theta \mid X_1, \ldots, X_n \sim \mathcal{N}\left(\left(\frac{n\tau_0}{n\tau_0 + \tau_{\text{prior}}}\right)\bar{X} + \left(\frac{\tau_{\text{prior}}}{n\tau_0 + \tau_{\text{prior}}}\right)\theta_0, \left(n\tau_0 + \tau_{\text{prior}}\right)^{-1}\right). \tag{85}$$

The Bayesian Approach to Parameter Estimation

Asymptotic Normality of the Posterior

- As $n \to \infty$, the posterior distribution near MLE can be approximated as a Gaussian. Let $\theta = \hat{\theta}_n^{\text{MLE}} + h/\sqrt{n}$ for some $h > 0$. Under some regularity conditions, with high probability,

$$
\begin{aligned}
\log f_{\Theta|\boldsymbol{X}}(\theta \mid \boldsymbol{x}) &= \log C_1(\boldsymbol{x}) + \log f_{\boldsymbol{X}|\Theta}(\boldsymbol{x} \mid \theta) + \log f_\Theta(\theta) \\
&= \log C_1(\boldsymbol{x}) + \Bigg(l_n(\hat{\theta}_n^{\text{MLE}}) + \underbrace{l_n'(\hat{\theta}_n^{\text{MLE}})}_{=0}(\theta - \hat{\theta}_n^{\text{MLE}}) + \frac{l_n''(\hat{\theta}_n^{\text{MLE}})}{2}(\theta - \hat{\theta}_n^{\text{MLE}})^2 \\
&\quad + \frac{l_n^{(3)}(\tilde{\theta}_{n,1})}{6}(\theta - \hat{\theta}_n^{\text{MLE}})^3 \Bigg) + \left(\log f_\Theta(\hat{\theta}_n^{\text{MLE}}) + \frac{f_\Theta'(\tilde{\theta}_{n,2})}{f_\Theta(\tilde{\theta}_{n,2})}(\theta - \hat{\theta}_n^{\text{MLE}}) \right) \\
&= \log C_2(\boldsymbol{x}) + \frac{1}{2}\bar{l}_n''(\hat{\theta}_n^{\text{MLE}})h^2 + r_n
\end{aligned}
\tag{86}
$$

where $r_n = \frac{\bar{l}_n^{(3)}(\tilde{\theta}_{n,1})h^3}{6\sqrt{n}} + \frac{f_\Theta'(\tilde{\theta}_{n,2})h}{f_\Theta(\tilde{\theta}_{n,2})\sqrt{n}} \xrightarrow{p} 0$ and $\tilde{\theta}_{n,1}$ and $\tilde{\theta}_{n,2}$ are between θ and $\hat{\theta}_n^{\text{MLE}}$.

Asymptotic Normality of the Posterior

- Therefore, for large n, the posterior distribution is approximately:

$$f_{\Theta|\boldsymbol{X}}(\theta \mid \boldsymbol{x}) \propto \exp\left(-\frac{(\theta - \hat{\theta}_n^{\mathrm{MLE}})^2}{2\left(-n\bar{l}_n''(\hat{\theta}_n^{\mathrm{MLE}})\right)^{-1}}\right), \quad |\theta - \hat{\theta}_n^{\mathrm{MLE}}| < \frac{h}{\sqrt{n}} \tag{87}$$

- Equivalently, this corresponds to the Normal distribution:

$$\Theta \mid \boldsymbol{X} \overset{\text{approx}}{\sim} \mathcal{N}\left(\hat{\theta}_n^{\mathrm{MLE}}, \left(-l_n''(\hat{\theta}_n^{\mathrm{MLE}})\right)^{-1}\right). \tag{88}$$

Table of Contents

1. Introduction
2. Fitting the Poisson Distribution to Emissions of Alpha Particles
3. Parameter Estimation
4. The Method of Moments
5. The Method of Maximum Likelihood
6. The Bayesian Approach to Parameter Estimation
7. Efficiency and the Cramér-Rao Lower Bound
8. Sufficiency

Relative Efficiency

- We have studied various estimators. How can we compare their efficiencies?
- For a given estimator $\hat{\theta}$ of θ_0, the mean squared error (MSE) is one of the basic criteria:

$$\mathrm{MSE}(\hat{\theta}) = \mathsf{E}_{\theta_0}[(\hat{\theta} - \theta_0)^2] = \mathsf{Var}_{\theta_0}[\hat{\theta}] + \left(\mathsf{E}_{\theta_0}[\hat{\theta}] - \theta_0\right)^2 \tag{89}$$

- When estimators are unbiased (i.e., $\mathsf{E}_{\theta_0}[\hat{\theta}] = \theta_0$), the MSE reduces to the variance of the estimator.

Relative Efficiency

- **Relative Efficiency**: For any two unbiased estimators $\hat{\theta}$ and $\tilde{\theta}$, the *relative efficiency* of $\hat{\theta}$ with respect to $\tilde{\theta}$ is defined as:

$$\mathrm{eff}(\hat{\theta}, \tilde{\theta}) = \frac{\mathrm{Var}_\theta[\hat{\theta}]}{\mathrm{Var}_\theta[\tilde{\theta}]} \tag{90}$$

- For example, let $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Poi}(\lambda)$. Consider two estimators: $\hat{\lambda}^{\mathrm{MLE}} = \hat{m}_1$ and $\hat{\lambda}^{\mathrm{MME}} = \hat{m}_2 - \hat{m}_1^2$. Their asymptotic relative efficiency is:

$$\lim_{n\to\infty} \frac{\mathrm{Var}_\lambda[\hat{\lambda}^{\mathrm{MLE}}]}{\mathrm{Var}_\lambda[\hat{\lambda}^{\mathrm{MME}}]} = \frac{1}{2\lambda + 1}. \tag{91}$$

Cramér-Rao Lower Bound

Exercise 18

Theorem 8.7.A (Cramér-Rao Lower Bound): Let $X_1, \ldots, X_n$ be i.i.d. with PDF $f(x;\theta)$. Let $T = t(X_1, \ldots, X_n)$ be an unbiased estimator of θ. Show that, under regularity conditions for the asymptotic normality of MLEs,

$$\mathrm{Var}_\theta[T] \geq \frac{1}{\mathcal{I}_n(\theta)} \tag{92}$$

Cramér-Rao Lower Bound

Cramér-Rao Lower Bound

- Based on the Cramér-Rao lower bound, the variance of any unbiased estimator cannot be smaller than $\mathcal{I}_n(\theta)^{-1}$.
- Therefore, if an unbiased estimator $\hat{\theta}^{\mathrm{UE}}$ has a variance equal to $\mathcal{I}_n(\theta)^{-1}$, then it is the *Minimum Variance Unbiased Estimator*.
- Recall the asymptotic normality of MLEs:

$$\sqrt{n}\left(\hat{\theta}_n^{\mathrm{MLE}} - \theta\right) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_1(\theta)^{-1}). \tag{93}$$

  This implies that for large n,

$$\mathrm{Var}_\theta\left[\hat{\theta}_n^{\mathrm{MLE}}\right] \approx \frac{1}{n\mathcal{I}_1(\theta)} = \frac{1}{\mathcal{I}_n(\theta)}.$$

- Under regularity conditions, if the MLE is unbiased, it is the most efficient. Even if the MLE is biased, note that MLEs are asymptotically unbiased, i.e., $\lim_{n\to\infty} \mathrm{E}_\theta[\hat{\theta}_n^{\mathrm{MLE}}] = \theta$.

Table of Contents

1. Introduction
2. Fitting the Poisson Distribution to Emissions of Alpha Particles
3. Parameter Estimation
4. The Method of Moments
5. The Method of Maximum Likelihood
6. The Bayesian Approach to Parameter Estimation
7. Efficiency and the Cramér-Rao Lower Bound
8. Sufficiency

Sufficiency

- So far, we have examined the asymptotic relative efficiency of various estimators, which compares MSEs when the sample size n is large.
- Although it is a principled criterion, it has limitations when n is not sufficiently large.
- Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\theta_0, 1)$. Consider a naive estimator $\hat{\theta}^{\text{Naive}} := c$ for some constant c. Then, $\mathrm{MSE}(\hat{\theta}^{\text{Naive}}) < \mathrm{MSE}(\hat{\theta}^{\text{MLE}})$ if and only if:

$$n < (\theta_0 - c)^{-2}. \tag{94}$$

- *Sufficiency* is another principled criterion in statistics for examining the quality of estimators, particularly valid for any sample size.

Sufficiency

- As a toy example, let X_1 and X_2 be i.i.d. Bernoulli random variables with parameter θ.
- When we observe $X_1 + X_2 = 1$, the possible outcomes are $(X_1, X_2) = (1, 0)$ or $(0, 1)$. In both cases, for any $\theta \in \Omega$:

$$p_{X_1,X_2|X_1+X_2}(1, 0 \mid 1; \theta) = \frac{\theta(1-\theta)}{\binom{2}{1}\theta(1-\theta)} = \frac{1}{2} \tag{95}$$

and

$$p_{X_1,X_2|X_1+X_2}(0, 1 \mid 1; \theta) = \frac{\theta(1-\theta)}{\binom{2}{1}\theta(1-\theta)} = \frac{1}{2}. \tag{96}$$

- Thus, the distribution of $(X_1, X_2)^T \mid X_1 + X_2 = 1$ does not depend on θ. Consequently, the conditional distribution of any statistic $g(X_1, X_2)$ given $X_1 + X_2 = 1$ is independent of θ.

Sufficient Statistic

- **Sufficient Statistic**: A statistic $T(X_1, \ldots, X_n)$ is said to be *sufficient* for θ if the conditional distribution of $X_1, \ldots, X_n$ given $T = t$ does not depend on θ for any value of t, equivalently,

$$p_{X_1,\ldots,X_n|T}(x_1, \ldots, x_n \mid t; \theta_1) = p_{X_1,\ldots,X_n|T}(x_1, \ldots, x_n \mid t; \theta_2) \quad \text{for any } \theta_1, \theta_2. \tag{97}$$

- Once we condition on T, thereby reflecting all information contained in it, the data $\boldsymbol{X}$ provide no further information to distinguish θ_1 from θ_2.

Sufficient Statistic

Exercise 19

Example 8.8.A: Let $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Ber}(\theta)$. Show that $T = \sum_{i=1}^{n} X_i$ is a sufficient statistic for θ.

Sufficient Statistic

Exercise 20

Theorem 8.8.1.A (Factorization Theorem): Show that, a statistic $T(X_1, \ldots, X_n)$ is sufficient for θ if and only if the joint PMF[a] can be expressed as:

$$p_{X_1,\ldots,X_n}(x_1, \ldots, x_n; \theta) = g\left(T(x_1, \ldots, x_n), \theta\right) h(x_1, \ldots, x_n) \tag{98}$$

[a] This result also holds for the continuous case.

Sufficient Statistic

Sufficient Statistic

Exercise 21

Example 8.8.1.B: Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ where $\theta = (\mu, \sigma^2)^T$. Show that $\boldsymbol{T}(\boldsymbol{X}) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2\right)^T$ is a sufficient statistic for θ.

Sufficient Statistic

- Note that when $\boldsymbol{X}$ follows a multi-parameter exponential family distribution, i.e.,

$$f_{\boldsymbol{X}}(\boldsymbol{x};\boldsymbol{\eta}) = h(\boldsymbol{x}) \exp\left(\boldsymbol{\eta}^\top \boldsymbol{T}(\boldsymbol{x}) - A(\boldsymbol{\eta})\right), \tag{99}$$

where $\boldsymbol{\eta} = \boldsymbol{\eta}(\theta)$ is the vector of natural parameters, $\boldsymbol{T}(\boldsymbol{x})$ is the vector of sufficient statistics, and $A(\boldsymbol{\eta})$ is the log-partition function, then

$$\boldsymbol{T}(\boldsymbol{X}) \tag{100}$$

is a sufficient statistic for $\boldsymbol{\eta}$.

The Rao-Blackwell Theorem

- **Corollary 8.8.1.A**: If $T(\boldsymbol{X})$ is sufficient for θ, the MLE depends on the data $\boldsymbol{X}$ only through T.
- We can utilize sufficient statistics to improve arbitrary estimators.
- **Theorem 8.8.2.A (The Rao-Blackwell Theorem)**: Let $T(\boldsymbol{X})$ be a sufficient statistic for θ and let $\hat{\theta}$ be an arbitrary estimator with $\mathrm{E}[\hat{\theta}^2] < \infty$. Define $\hat{\theta}^{\mathrm{RB}} = \mathrm{E}[\hat{\theta} \mid T(\boldsymbol{X})]$. Then,

$$\mathrm{MSE}(\hat{\theta}^{\mathrm{RB}}) \leq \mathrm{MSE}(\hat{\theta}), \tag{101}$$

where the equality holds if and only if $\hat{\theta}^{\mathrm{RB}} = \hat{\theta}$.

The Rao-Blackwell Theorem

Exercise 22

Prove the Rao-Blackwell Theorem: Let $T(\boldsymbol{X})$ be a sufficient statistic for θ and let $\hat{\theta}$ be an arbitrary estimator with $\mathsf{E}[\hat{\theta}^2] < \infty$. Define $\hat{\theta}^{\mathsf{RB}} = \mathsf{E}[\hat{\theta} \mid T(\boldsymbol{X})]$. Then,

$$\mathrm{MSE}(\hat{\theta}^{\mathsf{RB}}) \leq \mathrm{MSE}(\hat{\theta}), \tag{102}$$

where the equality holds if and only if $\hat{\theta}^{\mathsf{RB}} = \hat{\theta}$.

The Rao-Blackwell Theorem

Practice Problem I

1. An estimator $\hat{\theta}_n$ for a parameter θ is called *asymptotically unbiased* if $\lim_{n\to\infty} \mathrm{E}[\hat{\theta}_n] = \theta$. Show that if an estimator $\hat{\theta}_n$ is asymptotically unbiased and $\lim_{n\to\infty} \mathrm{Var}[\hat{\theta}_n] = 0$, then the estimator is consistent.
2. Let $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} \mathrm{Bin}(n, p)$, where n is known, $\theta = p$, and $\Omega = [0, 1]$. Find the MLE for p.
3. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{U}(a, b)$, where $\theta = (a, b)^T$ and $\Omega = \{(a, b) : -\infty < a < b < \infty\}$. Find the MLE for θ.
4. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{U}(\theta - 1, \theta + 1)$, where $\Omega = (-\infty, \infty)$. Find all MLEs for θ.

Practice Problem II

5. **Double Exponential (Laplace)**: Let $X_1, \ldots, X_{2m-1} \overset{\text{i.i.d.}}{\sim} \mathrm{Laplace}(\mu, b)$ where $m \in \mathbb{N}$. Its PDF is given by:

$$f(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right),$$

where $\theta = (\mu, b)^T$ and $\Omega = \{(\mu, b)^T : -\infty < \mu < \infty, 0 < b\}$. Find the MLE for $\theta = (\mu, b)^T$.

6. **Pareto Distribution**: Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{Pareto}(\alpha, \beta)$. Its PDF is given by:

$$f(x; \alpha, \beta) = \frac{\alpha\beta^\alpha}{x^{\alpha+1}}, \quad x \geq \beta,$$

and zero otherwise. Here, $\theta = (\alpha, \beta)^T$ and $\Omega = \{(\alpha, \beta) : \alpha > 0, \beta > 0\}$. Find the MLE for θ.

Practice Problem III

7. Let $\boldsymbol{X}$ follow a multi-parameter exponential family distribution whose PDF is given by:

$$f_{\boldsymbol{X}}(\boldsymbol{x};\boldsymbol{\eta}) = h(\boldsymbol{x}) \exp\left(\boldsymbol{\eta}^T \boldsymbol{T}(\boldsymbol{x}) - A(\boldsymbol{\eta})\right), \tag{103}$$

satisfying the following regularity conditions:

(C1) The natural parameter space $\boldsymbol{\eta}(\Omega)$ is an open set in $\mathbb{R}^k$.

(C2) The covariance matrix $\mathrm{Var}_{\boldsymbol{\eta}}[\boldsymbol{T}(\boldsymbol{X})]$ is positive definite (non-singular) for all $\boldsymbol{\eta} \in \boldsymbol{\eta}(\Omega)$.

a) Show that $\nabla A(\boldsymbol{\eta}) = \mathrm{E}_{\boldsymbol{\eta}}[\boldsymbol{T}(\boldsymbol{X})]$ and $\nabla^2 A(\boldsymbol{\eta}) = \mathrm{Var}_{\boldsymbol{\eta}}[\boldsymbol{T}(\boldsymbol{X})]$.

b) Show that $\hat{\boldsymbol{\eta}} \in \boldsymbol{\eta}(\Omega)$ is the unique MLE if it satisfies the following likelihood equation:

$$\mathrm{E}_{\hat{\boldsymbol{\eta}}}[\boldsymbol{T}(\boldsymbol{X})] = \overline{\boldsymbol{T}(\boldsymbol{X})} := n^{-1} \sum_{i=1}^{n} \boldsymbol{T}(\boldsymbol{X}_i). \tag{104}$$

Practice Problem IV

8. **Logistic Distribution with Location Parameter**: Let $X_1, \ldots, X_n$ be i.i.d. random variables with the following PDF:

$$f(x;\theta) = \frac{e^{-(x-\theta)}}{\left(1+e^{-(x-\theta)}\right)^2}.$$

Find the Fisher Information $\mathcal{I}_n(\theta)$.

9. **One-step Newton-Raphson Estimator**: Assume that all regularity conditions for the asymptotic normality of MLEs hold. Suppose that $\sqrt{n}\left(\hat{\theta}_n^{(0)} - \theta_0\right) \xrightarrow{d} Z$ for some random variable Z (e.g., MMEs). Define the one-step estimator as:

$$\hat{\theta}_n^{(1)} = \hat{\theta}_n^{(0)} + \left[-\bar{l}_n''(\hat{\theta}^{(0)})\right]^{-1} \bar{l}_n'(\hat{\theta}^{(0)}). \tag{105}$$

Show that $\hat{\theta}_n^{(1)}$ has the same asymptotic distribution as the MLE.

Practice Problem V

10. Let $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Beta}(\alpha, 1)$. Consider the following estimator for α:

$$\hat{\alpha}_n = \frac{n-1}{-\sum_{i=1}^{n} \log X_i}. \tag{106}$$

   a) Show that $-\alpha \sum_{i=1}^{n} \log X_i \sim \mathrm{Gamma}(n, 1)$.
   b) Show that $\hat{\alpha}_n$ is an unbiased estimator of α.
   c) Verify that $\hat{\alpha}_n$ is asymptotically efficient by showing:

$$\lim_{n\to\infty} \frac{\mathrm{Var}_\alpha[\hat{\alpha}_n]}{\left(n\mathcal{I}_1(\alpha)\right)^{-1}} = 1. \tag{107}$$

11. Let $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Gamma}(\alpha, \lambda)$ where $\theta = (\alpha, \lambda)^T$. Show that $\boldsymbol{T}(\boldsymbol{X}) = \left(\sum_{i=1}^{n} X_i, \prod_{i=1}^{n} X_i\right)^T$ is a sufficient statistic for θ.

Practice Problem VI

12. Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{U}(l, u)$ where $\theta = (l, u)^T$. Show that $\boldsymbol{T}(\boldsymbol{X}) = \left(X_{(1)}, X_{(n)}\right)^T$ is a sufficient statistic for θ.
13. **Linear Regression**: Let $(X_1, X_2, Y)^T$ follow a *multiple linear regression* model:

$$f_{(X_1,X_2,Y)}(x_1, x_2, y; \theta) = f_{\boldsymbol{X}}(\boldsymbol{x})\mathcal{N}(y; \beta_0 + \beta_1 x_1 + \beta_2 x_2, \sigma^2) \tag{108}$$

where $f_{\boldsymbol{X}}(\boldsymbol{x})$ does not depend on θ and is non-degenerate, i.e., $\text{Var}[\boldsymbol{X}]$ is positive definite, and $\theta = (\beta_0, \beta_1, \beta_2, \sigma^2)^T$.

a) Find the MLE for θ, $\hat{\theta}^{\text{MLE}}$.

Practice Problem VII

b) Let $\tilde{\boldsymbol{X}}$ be the $n \times 3$ observed *design matrix*:

$$\tilde{\boldsymbol{X}} = \begin{pmatrix} 1 & x_{11} & x_{12} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{pmatrix}. \tag{109}$$

Find the conditional variance $\mathrm{Var}[(\hat{\beta}_0^{\mathrm{MLE}}, \hat{\beta}_1^{\mathrm{MLE}}, \hat{\beta}_2^{\mathrm{MLE}})^T \mid \tilde{\boldsymbol{X}}]$.

c) Show that $\mathrm{Var}[\hat{\beta}_1^{\mathrm{MLE}} \mid \tilde{\boldsymbol{X}}] \to \infty$ and $\mathrm{Var}[\hat{\beta}_2^{\mathrm{MLE}} \mid \tilde{\boldsymbol{X}}] \to \infty$ as $|r_{X_1, X_2}| \to 1$.

Practice Problem VIII

14. **Logistic Regression**: Let $(X, Y)^T$ follow a *logistic model (or logit model)*:

$$f_{(X,Y)}(x, y; \theta) = f_X(x)\mathrm{Ber}(y; p(x; \theta)) \tag{110}$$

where $\mathrm{Ber}(y; p)$ denotes the PMF of a Bernoulli distribution with parameter p, $\theta = (\beta_0, \beta_1)^T$, and $f_X(x)$ is the marginal distribution of X, which does not depend on θ. The success probability is defined by the logistic function:

$$p(x; \theta) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 x))}. \tag{111}$$

a) Show that the logit model in Equation (111) is equivalent to:

$$\log \frac{p(x; \theta)}{1 - p(x; \theta)} = \beta_0 + \beta_1 x. \tag{112}$$

Practice Problem IX

b) Let $\tilde{\boldsymbol{X}}$ be the $n \times 2$ observed *design matrix*:

$$\tilde{\boldsymbol{X}} = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}. \tag{113}$$

Show that $\mathcal{I}_n(\theta \mid \tilde{\boldsymbol{X}}) := \mathrm{Var}_\theta[\nabla_\theta l_n(\theta) \mid \tilde{\boldsymbol{X}}]$ is given by:

$$\mathcal{I}_n(\theta) = \tilde{\boldsymbol{X}}^T \boldsymbol{W} \tilde{\boldsymbol{X}} \tag{114}$$

where $\boldsymbol{W}$ is an $n \times n$ diagonal matrix with elements $W_{ii} = p(x_i; \theta)(1 - p(x_i; \theta))$.

Practice Problem X

15. **Multi-class Logistic Regression**: Let $(X, Y_1, Y_2, Y_3)^T$ follow a *multi-class logistic model*:

$$f_{(X,\boldsymbol{Y})}(x, \boldsymbol{y}; \theta) = f_X(x)\mathrm{Cat}(\boldsymbol{y}; \boldsymbol{p}(x; \theta)) \tag{115}$$

where $\boldsymbol{y} \in \{(1,0,0)^T, (0,1,0)^T, (0,0,1)^T\}$, $x \in \mathbb{R}$, and $\mathrm{Cat}(\boldsymbol{y}; \boldsymbol{p})$ denotes the PMF of a categorical distribution with parameter $\boldsymbol{p} = (p_1, p_2, p_3)^T \in \{\boldsymbol{p} : p_k \geq 0, \sum_{k=1}^3 p_k = 1\}$:

$$\mathrm{Cat}(\boldsymbol{y}; \boldsymbol{p}) = \begin{cases} p_1^{y_1} p_2^{y_2} p_3^{y_3} & \text{if } y_k \in \{0,1\} \text{ for } k = 1,2,3 \text{ and } \sum_{k=1}^3 y_k = 1, \\ 0 & \text{otherwise.} \end{cases} \tag{116}$$

Practice Problem XI

Here, $f_X(x)$ is the marginal distribution of X, which does not depend on θ. Assuming class 3 is the reference category, let $\boldsymbol{\beta}_1 = (\beta_{10}, \beta_{11})^T$ and $\boldsymbol{\beta}_2 = (\beta_{20}, \beta_{21})^T$, so that $\theta = (\boldsymbol{\beta}_1^T, \boldsymbol{\beta}_2^T)^T$. The success probability vector is defined by the *softmax function*:

$$p_k(x;\theta) = \frac{\exp(\beta_{k0} + \beta_{k1}x)}{1 + \sum_{j=1}^{2} \exp(\beta_{j0} + \beta_{j1}x)} \quad \text{for } k = 1, 2, \tag{117}$$

and $p_3(x;\theta) = \frac{1}{1+\sum_{j=1}^{2} \exp(\beta_{j0}+\beta_{j1}x)}$.

a) Show that the multi-class model in Equation (117) is equivalent to the following log-odds system:

$$\log \frac{p_k(x;\theta)}{p_3(x;\theta)} = \beta_{k0} + \beta_{k1}x \quad \text{for } k = 1, 2. \tag{118}$$

Practice Problem XII

b) Let $\tilde{\boldsymbol{X}}$ be the $n \times 2$ observed *design matrix*:

$$\tilde{\boldsymbol{X}} = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}. \tag{119}$$

Show that the score vector with respect to $\boldsymbol{\beta}_k$ $(k = 1, 2)$ is given by:

$$\nabla_{\boldsymbol{\beta}_k} l_n(\theta) = \tilde{\boldsymbol{X}}^T (\boldsymbol{Y}_k - \boldsymbol{p}_k) \tag{120}$$

where $\boldsymbol{Y}_k = (y_{1k}, \ldots, y_{nk})^T$ and $\boldsymbol{p}_k = (p_k(x_1; \theta), \ldots, p_k(x_n; \theta))^T$.

Practice Problem XIII

16. **Autoregressive Model of Order 1**: Let $X_1, \ldots, X_T$ ($T \geq 3$) be a time series generated from an *autoregressive model of order 1* (AR(1)):

$$X_t = \rho X_{t-1} + \epsilon_t, \quad t = 2, \ldots, T, \tag{121}$$

where $\epsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$. The joint probability density function is given by:

$$\begin{aligned} f(\boldsymbol{x}; \theta) &= f_{X_1}(x_1) \prod_{t=2}^{T} f_{X_t | X_1, \ldots, X_{t-1}}(x_t \mid x_1, \ldots, x_{t-1}; \theta) \\ &= f_{X_1}(x_1) \prod_{t=2}^{T} \mathcal{N}(x_t; \rho x_{t-1}, \sigma^2) \end{aligned} \tag{122}$$

where $\theta = (\rho, \sigma^2)^T$ and $f_{X_1}(x_1)$ does not depend on θ.

Practice Problem XIV

a) Find $\mathrm{Cov}[X_{t-1}, X_t \mid X_1, \ldots, X_{t-2}]$ where $t = 3, \ldots, T$.
b) Find the MLE for θ, $\hat{\theta}^{\mathrm{MLE}}$.
c) Show that $\mathcal{I}_{T-1}(\theta \mid X_1, \ldots, X_{T-1}) := \mathrm{Var}_\theta \left[\nabla_\theta \log f(X_T \mid X_1, \ldots, X_{T-1}; \theta) \mid X_1, \ldots, X_{T-1}\right]$ is given by:

$$\mathcal{I}_{T-1}(\theta \mid X_1, \ldots, X_{T-1}) = \frac{1}{\sigma^2} \begin{pmatrix} X_{T-1}^2 & 0 \\ 0 & 1/2 \end{pmatrix}. \tag{123}$$

Advanced Practice Problem I

1. a) Let $X_1, \ldots, X_n$ be i.i.d. samples with a finite fourth moment. Show that:

$$\sqrt{n}\left(\frac{\hat{\sigma^2}^{\text{MME}}}{\sigma^2} - 1\right) \xrightarrow{d} Z, \quad Z \sim \mathcal{N}\left(0, \mathrm{E}\left[\left(\frac{X - \mu_X}{\sigma_X}\right)^4\right] - 1\right), \tag{124}$$

where $\hat{\sigma^2}^{\text{MME}} = n^{-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 = \hat{m}_2 - \hat{m}_1^2$.

b) When $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Poi}(\lambda)$, we can consider two MMEs: $\hat{\lambda}_1^{\text{MME}} = \hat{m}_1$ and $\hat{\lambda}_2^{\text{MME}} = \hat{m}_2 - \hat{m}_1^2$. Given $\mathrm{E}[(X_1 - \lambda)^4] = \lambda(1 + 3\lambda)$, show that

$$\lim_{n\to\infty} \frac{\mathrm{Var}_\lambda[\hat{\lambda}_1^{\text{MME}}]}{\mathrm{Var}_\lambda[\hat{\lambda}_2^{\text{MME}}]} = \frac{1}{2\lambda + 1}. \tag{125}$$

Advanced Practice Problem II

2. a) Let $(X_n)_{n\geq 1}$ be a sequence of random variables and θ be a constant. Suppose that $\sqrt{n}\,(X_n - \theta) \xrightarrow{d} Z$ for some random variable Z. Then, for any twice differentiable function g with a continuous second derivative such that $g'(\theta) = 0$, show that:

$$n\left(g(X_n) - g(\theta)\right) \xrightarrow{d} \frac{g''(\theta)}{2} Z^2. \tag{126}$$

b) Let X_i be i.i.d. standardized random variables (i.e., $\mu = 0$ and $\sigma^2 = 1$). Show that, for any $0 < \alpha < 0.5$:

$$\lim_{n\to\infty} P\left(1 - \frac{1}{2n}\chi^2_{\alpha/2}(1) \leq \cos \bar{X} \leq 1 - \frac{1}{2n}\chi^2_{1-\alpha/2}(1)\right) = 1 - \alpha. \tag{127}$$

Advanced Practice Problem III

3. Suppose $(X_i, Y_i)^\top$ for $i = 1, \ldots, n$ are i.i.d. random vectors from a bivariate normal distribution with parameter $\theta = (\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)^\top$. The parameter space is given by:

$$\Omega = \{\theta : \mu_X, \mu_Y \in \mathbb{R}, \sigma_X^2, \sigma_Y^2 > 0, -1 < \rho < 1\}.$$

   a) Show that the MLE over $\hat{\theta}$ is given by:

$$\hat{\theta} = \left(\bar{X}, \bar{Y}, \frac{(n-1)S_X^2}{n}, \frac{(n-1)S_Y^2}{n}, R\right)^T, \tag{128}$$

   where R denotes the sample correlation coefficient.

   b) Derive the asymptotic distribution of $\sqrt{n}\,(\hat{\rho} - \rho)$.

   c) Find a variance-stabilizing transformation $g(\rho)$ such that the asymptotic distribution of $\sqrt{n}\left(g(\hat{\rho}) - g(\rho)\right)$ is $\mathcal{N}(0, 1)$.

Advanced Practice Problem IV

4. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{U}(0, \theta)$.
   a) Show that $\mathrm{E}[X_{(r)} \mid X_{(n)}] = \frac{r}{n} X_{(n)}$ for $1 \leq r \leq n$.
   b) Let $\hat{\theta}^{\mathrm{RB}} = \mathrm{E}[2\bar{X} \mid X_{(n)}]$. Show that

$$\mathrm{MSE}(\hat{\theta}^{\mathrm{RB}}) = \frac{\theta^2}{n(n+2)}. \tag{129}$$

5. **Expectation-Maximization**: Suppose we observe data $\boldsymbol{X} = (X_1, \ldots, X_n)^T$ from a two-component Gaussian Mixture Model:

$$\begin{aligned} &Z_1, \ldots, Z_n \overset{\text{i.i.d.}}{\sim} \mathrm{Ber}(\pi) \\ &X_i \mid \boldsymbol{Z} \sim \mathcal{N}(\mu_{Z_i}, \sigma^2_{Z_i}) \quad \text{for } i = 1, \ldots, n \end{aligned} \tag{130}$$

where $\boldsymbol{Z}$ is latent, $X_1, \ldots, X_n$ are (conditionally) independent given $\boldsymbol{Z}$, and $\theta = (\pi, \mu_0, \mu_1, \sigma_0^2, \sigma_1^2)^T$.

Advanced Practice Problem V

a) E-step: Let the *complete-data log-likelihood* be $l_c(\theta; \boldsymbol{X}, \boldsymbol{Z}) = \log\left(f_\theta(\boldsymbol{X} \mid \boldsymbol{Z}) p_\theta(\boldsymbol{Z})\right)$. Let $\hat{\theta}^{(t)}$ be the parameter estimate at the t-th iteration. Define

$$Q(\theta \mid \hat{\theta}^{(t)}) := \mathrm{E}_{\boldsymbol{Z} \sim p(\boldsymbol{z} \mid \boldsymbol{X}; \hat{\theta}^{(t)})}\left[l_c(\theta; \boldsymbol{X}, \boldsymbol{Z})\right]. \tag{131}$$

Letting $\gamma_{ik}^{(t)} := P(Z_i = k \mid X_i; \hat{\theta}^{(t)})$ for $k \in \{0, 1\}$, show that

$$Q(\theta \mid \hat{\theta}^{(t)}) = \sum_{i=1}^{n} \left(\gamma_{i1}^{(t)} \log \pi + \gamma_{i0}^{(t)} \log(1 - \pi) + \sum_{k=0}^{1} \gamma_{ik}^{(t)} \log \mathcal{N}(X_i; \mu_k, \sigma_k^2) \right). \tag{132}$$

Advanced Practice Problem VI

b) M-step: Define the updating rule as:

$$\hat{\theta}^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \hat{\theta}^{(t)}). \tag{133}$$

Show that

$$\hat{\pi}^{(t+1)} = \frac{1}{n}\sum_{i=1}^{n}\gamma_{i1}^{(t)}, \quad \hat{\mu}_k^{(t+1)} = \frac{\sum_{i=1}^{n}\gamma_{ik}^{(t)}X_i}{\sum_{i=1}^{n}\gamma_{ik}^{(t)}}, \quad (\hat{\sigma}_k^{2\,(t+1)}) = \frac{\sum_{i=1}^{n}\gamma_{ik}^{(t)}(X_i-\hat{\mu}_k^{(t+1)})^2}{\sum_{i=1}^{n}\gamma_{ik}^{(t)}}. \tag{134}$$

c) Show that the marginal log-likelihood $l(\theta; \boldsymbol{X}) = \sum_{i=1}^{n}\log f_\theta(X_i)$ never decreases, i.e.,

$$l(\hat{\theta}^{(t+1)}; \boldsymbol{X}) \geq l(\hat{\theta}^{(t)}; \boldsymbol{X}). \tag{135}$$

In this problem, you may use Jensen's inequality, $\log \mathrm{E}[Y] \geq \mathrm{E}[\log Y]$, without proof.

Advanced Practice Problem VII

6. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} f(x;\theta)$ and let $T(\boldsymbol{X})$ be a *sufficient statistic* for θ. Suppose T is also a *complete statistic*, meaning that for any (measurable) function g, the following condition holds:

$$\mathrm{E}_\theta[g(T)] = 0 \quad \text{for all } \theta \in \Omega \implies P_\theta(g(T) = 0) = 1 \quad \text{for all } \theta \in \Omega. \tag{136}$$

Show that, for any unbiased estimator $\hat{\theta}^{\mathrm{UE}}$ for θ, the Rao-Blackwell estimator based on the complete sufficient statistic T,

$$\hat{\theta}^{\mathrm{RB}} := \mathrm{E}_\theta[\hat{\theta}^{\mathrm{UE}} \mid T], \tag{137}$$

is a Uniformly Minimum-Variance Unbiased Estimator (UMVUE). That is, the variance of any unbiased estimator is greater than or equal to $\mathrm{Var}_\theta[\hat{\theta}^{\mathrm{RB}}]$ for all $\theta \in \Omega$.

References I

Rice, J. A. (2007). *Mathematical statistics and data analysis*, volume 371. Thomson/Brooks/Cole Belmont, CA.